Follow the Workers, Not the Bids: Cartel Detection through Employer–Employee Networks*

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Abstract

Brazil's antitrust authority convicted 35 firms across seven procurement cartels in São Paulo between 2009 and 2019. We link these firms to 3.8 million bids on the state's electronic platform and to the universe of matched employer–employee records, assembling what is, to our knowledge, the first detection benchmark to combine auction behavior, labor microdata, and formal antitrust rulings.

The benchmark exposes a sharp divide. On roughly 2,300 close-margin auctions involving at least one convicted firm, canonical variance screens behave as bid-rotation theory predicts—dispersion and bidder counts are compressed—yet the areas under the ROC curve (0.52–0.56) are too low to be operationally useful. The close-margin RD of Kawai et al. (2023) likewise yields a null. Worker-flow screens tell a different story: counting the employees who have moved between the winning and runner-up firms produces AUCs of 0.75 and 0.74, outperforming every classical screen by roughly 20 percentage points. A co-bidding density measure reaches 0.70. Crucially, the worker-flow advantage is not uniform: in thin labor markets the gap reaches 33 percentage points; in thick markets it narrows to 10. A specialization placebo shows that about half the signal reflects the natural personnel circulation of thin markets, while the other half is a cartel-specific residual.

The pattern points to an institutional logic. Where specialized-labor pools are shallow, the cheapest way to coordinate a cartel runs through people, not through prices—and the traces show up in employment records that canonical bid screens never see.

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For competition authorities with access to employer–employee microdata, worker-flow screens offer a practical, low-cost complement to the toolkit they already have.

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1. Introduction

Do the canonical tools for detecting procurement cartels work when they leave the thick-market, high-enforcement environments in which they were built? We put this question to a new test. Linking seven cartels convicted by Brazil's antitrust authority to 3.8 million bids and the full universe of formal employment relationships in São Paulo, we assemble the first detection benchmark that combines auction data, employer–employee records, and criminal ground truth. Classical variance screens and a close-margin regression discontinuity yield AUCs of 0.52–0.56—directionally right but operationally inadequate. Screens built from inter-firm worker flows reach 0.75, and the gap widens sharply in thin labor markets where collusion coordinates through people rather than prices. This paper documents these facts, traces them to an institutional mechanism, and draws out the consequences for enforcement design.

A growing literature uses statistical regularities in bidding behavior to infer collusion that is never documented directly. Classical screens exploit the compression that bid-rotation agreements impose on the bid distribution: prearranged prices cluster around the designated winner, yielding low dispersion, few active bidders, and predictable rotation (Porter and Zona, 1993; Pesendorfer, 2000; Bajari and Ye, 2003; Conley and Decarolis, 2016; Imhof et al., 2018). The complementary approach of Kawai et al. (2023) focuses on the winning–losing threshold: under genuine competition, firms that barely win and barely lose should look alike, and any sharp discontinuity at the cutoff identifies a non-competitive wedge. Both families of tests have been validated primarily against cartels in high-income economies with deep bidder pools and strong institutional capacity for enforcement; their track record in thinner settings rests on a small and growing empirical base.

Network-based methods have pushed the frontier in two directions. Within the procurement data itself, co-bidding networks flag candidate cartels through group-level cohesion and exclusivity (Wachs and Kertész, 2019; Signor et al., 2020; Huber and Imhof, 2019; Clark et al., 2018). Outside the bid record, a recent wave exploits external network layers: Asai and Charoenwong (2026) show that common-ownership links predict bid rigging in Singapore, and Villamil et al. (2024) build multiplex ownership–co-bidding networks for Swedish procurement. What remains missing is a network layer that captures *operational* coordination—the personnel flows, knowledge transfers, and shared specialized labor through which cartels in thin markets actually run—and a validation of such a

layer against criminal ground truth in a middle-income setting. The São Paulo–CADE–RAIS linkage we construct offers precisely that test, and is the first cartel-detection benchmark to integrate auction-level data, matched employer–employee records, and formal antitrust convictions. The policy stakes are immediate: middle-income competition authorities increasingly rely on screens developed and validated abroad to triage investigations (Organisation for Economic Co-operation and Development, 2015; Bosio et al., 2022), and if those screens systematically miss cartels under local conditions, enforcement pipelines may be misfiring where deterrence is needed most.

Our primary evaluation sample is a *cartel-active* benchmark of roughly 2,300 close-margin pregão auctions in which at least one convicted firm bid during the active period of its cartel (details of sample construction in Section 5).

On that benchmark, canonical screens disappoint. Bid dispersion and the coefficient of variation are weakly compressed at cartel-tainted auctions (Cohen's d between -0.07 and -0.24), and Welch t-statistics are significant thanks to the enormous control sample—but the AUCs top out at 0.56, far short of what a triage tool would require. The close-margin RD produces a null on the cartel subsample. In short, the screens pick up a whisper of signal, but not enough to act on.

Worker-flow screens recover what bid screens cannot. Counting the employees who have moved between the winner and the runner-up over the preceding five years yields an AUC of 0.75; the Jaccard similarity of the two firms' workforces reaches 0.74; a co-bidding density measure adds 0.70. A specialization placebo decomposes the worker-flow signal into a same-market component—reflecting the natural labor circulation of thin product markets—and a cartel-specific residual. Roughly half the signal comes from each channel, consistent with cartels both inheriting and reinforcing the personnel flows already present in the markets where they operate.

We read these findings through the lens of *institutional thinness*. When the universe of firms capable of bidding on a given product is small and the formal-sector labor pool that serves those firms is shallow, coordination through personnel becomes the cheapest technology available to a cartel. The bid distribution, in such markets, carries little trace of the collusion—but the employment records do. A simple model in Section 4.6 formalizes this logic and delivers three testable predictions: classical screens should be weakest in thin markets; worker-flow screens should be strongest there; and a same-market placebo should absorb part, but not all, of the worker-flow signal. All three predictions are borne out. The sharpest confirmation comes from splitting the bench-

mark by market thickness: the gap between the best classical screen and the worker-flow screen jumps from 10 percentage points in thick markets to 33 percentage points in thin ones.

The paper makes three contributions. The first is conceptual. Collusion leaves *institution-specific* signatures: the screen that detects a cartel depends on which coordination technology is cheapest in the relevant environment. In thick labor markets, collusion channels through bids and classical screens fire; in thin ones, it channels through people and worker-flow screens fire. This framing organizes—and partially reconciles—the contradictory findings that have accumulated as detection methods developed in one setting are deployed in another (Conley and Decarolis, 2016; Imhof et al., 2018; Kawai et al., 2023). The second contribution is substantive. We build conviction-anchored benchmarks by linking CADE rulings, BEC bid-level records, and RAIS employer–employee data, and show that worker-flow and co-bidding screens constructed from these sources substantially outperform classical screens. Because matched employer–employee registries are increasingly available to national competition authorities in middle-income economies, the screens we propose are directly deployable at negligible marginal cost. The third contribution is methodological: we deliver the first stress test of canonical detection frameworks against criminal ground truth outside a high-income setting, documenting that classical screens and the Kawai RD yield AUCs in the low 0.50s while network screens reach 0.75.

The remainder of the paper proceeds as follows. Section 3 describes public procurement in São Paulo and the CADE cartel cases that anchor the ground truth. Section 2 situates our work in the detection, network, and worker-mobility literatures. Section 5 describes the data sources and sample construction. Section 4 presents the empirical strategy and a simple model of institution-dependent coordination. Section 6 reports the results. Section 7 discusses policy implications and concludes.

2. Related Literature

Our paper sits at the intersection of three literatures: the design and validation of cartel-detection screens, the use of network structure to identify collusion, and the economics of inter-firm worker mobility.

2.1. Detection through bid-pattern screens

The idea that bid-rotation agreements leave statistical footprints is now well established. Porter and Zona (1993) showed that collusive phases of the 1880s railroad freight cartel exhibit systematically lower bid dispersion than competitive phases, exploiting the cartel's own internal records as an exogenous regime switch. Pesendorfer (2000) documented bid rotation in Florida school-milk procurement; Bajari and Ye (2003) proposed exchangeability tests for bidder costs in Midwestern highway paving; Conley and Decarolis (2016) extended pair-level correlation tests to Italian procurement. Imhof et al. (2018) demonstrated that combining multiple screens in a portfolio outperforms any single indicator in Swiss data, and Abrantes-Metz et al. (2006) offered practical guidance on implementing variance screens while cautioning against single-indicator reliance. Harrington (2008) synthesized the design principles, emphasizing a point our findings will make concrete: the informativeness of any screen depends on the competitive counterfactual *and* the coordination technology the cartel actually uses.

A complementary strand focuses on the bid margin itself. Kawai et al. (2023) proposed a regression discontinuity centered on the winning–losing threshold in Japanese construction procurement: under genuine competition, narrowly winning and narrowly losing firms should look alike, and any sharp discontinuity identifies a non-competitive wedge. Chassang and Ortner (2019) developed a theoretical framework in which bid constraints endogenously shape the collusive technology—predicting, among other things, that format-specific screens may outperform generic variance tests, a prediction consistent with the format heterogeneity we document below.

A common thread across these contributions is that nearly all validation exercises rely on ground truth from high-income economies with deep bidder pools and well-resourced enforcement agencies. Levenstein and Suslow (2006) documented across 81 international cartels that market structure, transparency, and enforcement capacity jointly determine cartel success—suggesting that screens validated in one institutional regime need not travel well to another. Direct validation against criminal ground truth in a middle-income setting with structurally different labor markets remains scarce, and that is the gap our conviction-anchored benchmark is designed to fill.

2.2. Network-based and machine-learning approaches

A parallel literature has moved beyond univariate bid statistics to exploit the relational structure of procurement markets. Wachs and Kertész (2019) showed that co-

bidding networks in Georgian procurement flag candidate cartels through group-level cohesion and exclusivity, even without criminal ground truth. Signor et al. (2020) built a probabilistic detector from the data released by Brazil's *Operação Lava Jato*, reporting 81–96 percent accuracy on the Petrobras construction cartel. Huber and Imhof (2019) combined classical screens with supervised machine-learning classifiers in Swiss data, correctly predicting 84 percent of collusive tenders out of sample. Clark et al. (2018) documented how collusive networks in Québec construction co-evolve with political connections, illustrating the interplay between bid rigging and corruption.

A more recent wave extends the network logic to data sources *outside* the bid record. Asai and Charoenwong (2026) link shareholder registries to procurement auctions in Singapore and show that common-ownership links predict identical bids; removing the ownership-network effect would cut government spending by 4–6 percent. Villamil et al. (2024) build temporal multiplex networks that combine ownership ties and co-bidding patterns in Swedish procurement, finding that denser ownership connections predict higher single-bidding and more missing bidders. Together, these papers establish that external network layers—corporate ownership, in their case—carry detection signal that co-bidding statistics alone cannot reach.

We add a third layer: labor-market connectivity, constructed from matched employer–employee records. Worker-flow networks differ from ownership networks in two ways that matter for detection. They capture *operational* coordination—personnel rotation, knowledge transfer, shared technical staff—rather than financial ties, and thus speak to a distinct coordination channel. And they are available wherever governments maintain compulsory employer–employee registries (RAIS in Brazil, DADS in France, IAB in Germany), making the approach portable without new data collection. Because worker-flow data are informationally independent of both the bid record and the corporate registry, the risk of circularity between the screen and the behavior it targets is lower than for co-bidding or ownership-based approaches.

2.3. Worker mobility as an inter-firm channel

Labor economists have long recognized that workers carry knowledge between firms. Stoyanov and Zubanov (2012) showed that productivity gains spill across Danish firms through the mobility of their employees, with larger spillovers when the productivity gap between sender and receiver is wide. Serafinelli (2019) documented a similar pattern in Italy's Veneto region: workers arriving from high-productivity firms raise wages and

output at their new employers. Jarosch et al. (2021) developed an equilibrium model in which team composition and worker reallocation jointly shape firm-level human capital. The upshot is that worker flows are not just a labor-market phenomenon—they are an active conduit for inter-firm information transmission.

Three recent papers connect this conduit to collusion. Lamichhane and McGee (2024) provide experimental evidence that, in Bertrand markets, worker mobility between firms facilitates collusive pricing: workers carry pricing-strategy information, and firms that exchange employees coordinate on higher prices and more market sharing, especially in concentrated markets. Herrera-Caicedo et al. (2025) study product-market collusion facilitated by shared directors and executives. They find that common leadership raises the probability of collusion by 11 percentage points—and, strikingly, that inter-firm worker flows *fall* by roughly 20% once a board interlock is established. The substitution is telling: when collusion runs through leadership ties, personnel-flow coordination becomes redundant. Our institutional-thinness model predicts the mirror image: when leadership ties are absent and coordination relies on operational personnel exchange, worker-flow screens should carry the signal that ownership and board-interlock screens miss.

On the data-infrastructure side, two papers have already linked RAIS to Brazilian procurement—though not for detection. Barbosa and Straub (2020) track worker movements between public agencies and private suppliers, showing that supplier-to-administration revolving doors are detrimental to procurement efficiency. Szerman (2023) exploits a debarment-policy change to estimate the worker-level costs of anti-corruption enforcement, finding that blacklisted firms lose nearly half their employees. Both papers prove the RAIS–procurement linkage is feasible and productive; neither uses it as a cartel screen.

We bring these threads together. Lamichhane and McGee (2024) establish the mechanism—worker mobility as a collusion channel. Asai and Charoenwong (2026) and Villamil et al. (2024) show that external network layers carry detection signal. Barbosa and Straub (2020) and Szerman (2023) demonstrate the data infrastructure. What has been missing is the step from mechanism to screen: constructing a labor-network detection tool, validating it against formal convictions, and embedding it in a framework that predicts *when* worker-flow screens should outperform the alternatives. That is what this paper provides.

3. Institutional Background

3.1. Public procurement in São Paulo: the BEC platform

Public procurement in the state of São Paulo is conducted through a decentralized system of purchasing agencies, each referred to as a Public Buyer Unit (PBU). Since 2007, the use of the state's electronic procurement platform, the *Bolsa Eletrônica de Compras* (BEC), has been mandatory for PBUs when acquiring common goods and services. The BEC platform centralizes tender procedures and records detailed information on each procurement process. It plays a crucial role in São Paulo's procurement operations: for instance, in 2018 alone, approximately US$1.7 billion worth of goods and services were procured via BEC. Over the period from 2009 onward, the BEC system handled more than R$7.4 billion (approximately 34% of all state government purchases) for the State Health Secretariat (SES/SP) and other agencies in São Paulo. Given the large scale and value of these tenders, ensuring fairness and preventing collusion in the procurement process is of paramount importance.

The BEC catalog contains a comprehensive listing of standardized items and services, classified by category. Nearly 55,000 purchase offers were issued through BEC during our study period, covering about 6,350 distinct items ranging from medical supplies to office equipment. Each purchase offer (PO) is a digital solicitation document that identifies one or more items to be procured, along with the required quantities and specifications. Each item within a PO is auctioned separately; thus a unique procurement transaction can be defined by a combination of a PO number and an item code (POI). The BEC records include rich information for each POI, including the reference (reserve) price set by the buyer, the number of participating firms, all bids submitted (identifying winners and losers), the auction format used, and the identities of the buyer (PBU) and auctioneer.

São Paulo's PBUs predominantly use two types of competitive tendering procedures available on BEC: (i) a sealed-bid tender (*convite*) and (ii) a multi-round descending auction (*pregão eletrônico*). In a sealed-bid tender, firms submit confidential bids by a specified deadline. After the submission period closes, all bids are opened and the contract is awarded to the qualified bidder with the lowest bid price. The sealed-bid format is typically used for purchases up to a value limit of R$176,000 (approximately US$35,000). In contrast, the descending auction (pregão) has no value limit and involves an open, interactive bidding phase. Initially, firms submit sealed proposals which are

evaluated for qualification. Then, all qualified bids are announced (with the bidders' identities concealed) and an open auction phase commences. During this phase, which lasts for an initial 20 minutes, bidders can submit progressively lower bids, knowing the current lowest bid in real time. If a new bid is placed in the final minutes of this period (between 16 and 20 minutes), the auction is automatically extended by an additional 4 minutes. The auction concludes when no new bids arrive for 4 consecutive minutes. The winner is the bidder offering the lowest price at the end of this process, provided that price is below the government's preset reference price.

The key difference between the two formats lies in the opportunity for iterative bidding. The pregão format allows real-time bid competition and, after determining the lowest bid through the auction, a brief negotiation or confirmation step to potentially further reduce the price. In contrast, the convite is a single-shot process with no negotiation stage. From an oversight perspective, the open auction format provides more transparency during the bidding process (as all bids are observable in real time, albeit anonymously), whereas the sealed-bid format concentrates all transparency at the moment of opening. Both mechanisms are subject to anti-collusion regulations and audit processes, but their procedural differences can influence how collusion or favoritism might manifest. Our primary analysis focuses on pregão auctions, which account for the majority of BEC procurement by value and for which the close-margin regression-discontinuity framework is most naturally motivated; we report results on convite as a complementary robustness check.

3.2. Antitrust enforcement: CADE and the conviction-anchored ground truth

Brazil's federal competition authority, the *Conselho Administrativo de Defesa Econômica* (CADE), is an autonomous agency within the Ministry of Justice responsible for investigating and adjudicating anticompetitive conduct. CADE's Tribunal renders administrative decisions (not criminal convictions in the judicial sense) that carry binding fines and behavioral remedies. Its investigative arm, the *Superintendência-Geral* (SG), conducts preliminary inquiries and administrative proceedings, often drawing on leniency agreements (*Termos de Compromisso de Cessação*, TCCs) in which cartel participants cooperate in exchange for reduced penalties. The typical enforcement pipeline proceeds from an initial complaint or leniency application, through a formal administrative proceeding, to a Tribunal decision that either convicts or acquits the named firms.

Our ground truth consists of CADE administrative proceedings in which the Tribunal found firms guilty of cartel conduct in product markets relevant to procurement in the state of São Paulo. We identified these cases through a systematic search of CADE's public case repository, restricting attention to proceedings with final Tribunal decisions (not pending or preliminary-stage cases) that specify a cartel period and list named firms with identifiable CNPJs (the Brazilian taxpayer identification number for legal entities). The resulting set contains seven distinct cartels operating in São Paulo procurement markets between 2004 and 2019, spanning sectors as diverse as pharmaceuticals, school meals, trash bags, solar water heaters, and metropolitan transit. Together, these cartels involve 35 convicted firms whose 8-digit CNPJ roots can be matched to the BEC bidding records.

The critical feature of this ground truth for our detection exercise is that CADE's case files delineate both the *firms* and the *product-market-time windows* over which each ring was found to have operated. This allows us to label individual BEC auctions as cartel-tainted—not merely firms as cartel-affiliated—by matching each convicted firm's CNPJ root and the CADE-specified conduct period to the corresponding BEC tenders by item code and year. The resulting conviction-anchored benchmark thus incorporates both the identity and timing dimensions of collusion, rather than treating all auctions involving a flagged firm as equally suspicious.

4. Empirical Strategy

We deploy four complementary detection approaches on the conviction-anchored benchmark described in Section 5: a close-margin regression discontinuity (Section 4.1), classical bid-distribution screens (Section 4.2), network-based screens constructed from worker-flow and co-bidding data (Section 4.3), and a specialization placebo that decomposes the network signal into market-specialization and cartel-specific components (Section 4.5). We evaluate each screen's discriminative power using the area under the receiver-operating-characteristic curve (Section 4.4). Section 4.6 sketches a simple model linking the informativeness of each screen family to institutional primitives.

4.1. Close-margin regression discontinuity

Following Kawai et al. (2023), we exploit the winning–losing threshold in close-margin pregão auctions. Let MV_a denote the margin of victory in auction a, defined as the

percentage difference between the runner-up bid and the winning bid:

$$MV_a = \frac{b_a^{\text{runner-up}} - b_a^{\text{win}}}{b_a^{\text{win}}}.$$

The running variable is MV_a, with a cutoff at zero: auctions with $MV_a \approx 0$ are those in which the winner barely underbid the closest rival. Under competitive conditions, the observable pre-determined characteristics of narrowly winning and narrowly losing firms should be continuous at $MV_a = 0$. Any statistically sharp discontinuity at the cutoff identifies a non-competitive wedge—either collusion or external favoritism.

We estimate the discontinuity using local-polynomial regression with the bandwidth selector of Imhof et al. (2018). The two primary outcomes are *incumbency*—an indicator equal to one if the firm won the immediately preceding auction in the same market—and *last bidder*—an indicator for whether the firm submitted the final bid in the auction. A positive jump in incumbency at $MV = 0$ is consistent with bid rotation (the designated winner is systematically a past winner); a jump in last-bidder status is consistent with strategic bid timing. We supplement the RD estimates with a density test on the running variable, following Imhof et al. (2018), to assess whether there is manipulation of the bid margin itself—a prerequisite for interpreting discontinuities causally.

4.2. Classical bid-distribution screens

We implement four classical screens computed at the auction level on firm-best bids (one bid per firm, retaining the firm's lowest bid when multiple rounds exist):

1. **Coefficient of variation** (CV_a): the standard deviation of firm-best bids divided by their mean. Under bid rotation with cover bidding, CV_a is predicted to be lower at cartel-tainted auctions because bids cluster around the designated winner's price (Porter and Zona, 1993; Abrantes-Metz et al., 2006).
2. **Bid spread** (S_a): the range of firm-best bids normalized by their mean, $S_a = (\max - \min)/\text{mean}$. Like CV_a, the spread is predicted to be compressed under cover bidding.
3. **Margin of victory** (MV_a): defined above. Under bid rotation, the winning margin is predicted to be *small* because the designated winner undercuts cover bids by only the minimum necessary to win.

4. **Number of active bidders** (N_a): the count of distinct firms submitting bids. Collusive entry deterrence or market-division agreements predict *fewer* bidders at cartel-tainted auctions (Marshall and Marx, 2012).

Each screen is evaluated on the cartel-active benchmark ($|MV| \leq 0.10$), comparing the distribution of cartel-tainted auctions to clean controls. We report the Welch t-statistic for the mean difference, Cohen's d for effect-size calibration, the Wilcoxon rank-sum p-value for rank-order separation, and the AUC (Section 4.4).

4.3. Network-based screens

We construct three screens from administrative data sources external to the bid records themselves.

Worker-flow screens. For each auction-level (winner, runner-up) firm pair, we measure inter-firm labor connectivity using the five-year window of RAIS employer–employee records preceding the tender year. Two variants capture different aspects of labor proximity:

- **Shared workers** (W_{ij}): the number of distinct workers (identified by their PIS number) who appear in the employment records of both firm i and firm j at any point during the five-year window. A higher count indicates denser personnel flows between the two firms.

- **Jaccard similarity** (J_{ij}): the ratio $J_{ij} = W_{ij}/(N_i + N_j - W_{ij})$, where N_i and N_j are the total number of distinct workers employed by each firm over the window. The Jaccard index normalizes the shared-worker count by the size of each firm's workforce, ensuring that the screen is not mechanically driven by firm size.

The hypothesis is that cartel coordination leaves detectable traces in personnel flows: managers, salespeople, or technical staff who rotate between colluding firms serve as informal communication channels (Stoyanov and Zubanov, 2012; Serafinelli, 2019). Under the institutional-thinness mechanism we develop below, the worker-flow signal should be strongest in markets with thin specialized-labor pools.

Co-bidding screen. For each (winner, runner-up) pair, we count the number of auctions in which both firms submitted bids over the full sample period (C_{ij}). Firms that co-bid frequently may be coordinating market presence. The co-bidding screen provides

an independent network channel constructed entirely within the procurement data, serving as a comparator for the worker-flow screens, which are constructed from a wholly external source.

4.4. Detection evaluation: AUC

For each screen s, we evaluate discriminative power by computing the area under the receiver-operating-characteristic curve (AUC). The AUC is the probability that a randomly drawn cartel-tainted auction scores higher (more cartel-like) on screen s than a randomly drawn clean auction. For screens where lower values are predicted under collusion (CV, S, MV, N), we negate the scores before computing the AUC, so that a value above 0.5 always indicates detection in the predicted direction. An AUC of 0.5 corresponds to chance (no separation), and an AUC of 1.0 to perfect separation.

We compute the AUC via the Mann–Whitney U statistic with average-rank tie-handling:

$$\text{AUC} = \frac{U_1}{n_1 \times n_0},$$

where U_1 is the sum of ranks of cartel-tainted auctions minus $n_1(n_1+1)/2$, and n_1 and n_0 are the cartel and clean sample sizes, respectively. All screens—classical and network-based—are evaluated on the same unified close-margin benchmark ($|MV| \leq 0.10$), ensuring apples-to-apples comparability. Appendix A.1 reports AUCs at six alternative bandwidths (0.02–0.50); results are stable across the full range.

4.5. Specialization placebo

The worker-flow AUC may reflect two distinct channels: (i) *market specialization*—firms operating in the same thin product market share workers because the pool of specialized labor is small, regardless of whether they collude—and (ii) *cartel-specific coordination*—colluding firms engage in additional personnel transfers to sustain communication and monitoring. To disentangle these channels, we construct a three-way sample decomposition:

1. **Cartel**: auction-level (winner, runner-up) pairs in which at least one firm is cartel-active, drawn from the close-margin benchmark ($N \approx 2{,}300$ for pregão).
2. **Placebo (same market)**: non-cartel firm pairs bidding on the same BEC item code in the same year as each cartel-active auction. These pairs face the same

product-market labor pool as cartel pairs but are not colluding (by the ground-truth definition).

3. **Clean (global)**: all remaining non-cartel pairs in the close-margin benchmark.

If the entire worker-flow signal were driven by market specialization, the placebo pairs should exhibit the same AUC as cartel pairs (relative to the global clean baseline). If instead there is a cartel-specific residual, cartel pairs should exhibit *higher* worker-flow overlap than same-market placebo pairs. In a linear decomposition of the total AUC gain (cartel vs. clean minus 0.5), we attribute the fraction absorbed by the placebo to market specialization and the remainder to cartel-specific coordination.

4.6. A simple model of institution-dependent coordination

We sketch a parsimonious framework to organize the empirical predictions. Consider a market with $n \geq 2$ firms that can form a cartel. The cartel must coordinate on a winning bid for each auction. Two coordination technologies are available:

- **Bid-pattern coordination** (cost c_b): the cartel arranges bids so that cover bids cluster around the designated winner's price. This technology leaves traces in bid-distribution statistics (CV, spread, bidder counts) and is the mechanism assumed by classical screens.

- **Personnel-flow coordination** (cost $c_w(L)$, decreasing in the thickness L of the specialized-labor pool): the cartel coordinates through the transfer or rotation of key personnel—managers, salespeople, or technical staff—between member firms. This technology leaves traces in worker-flow statistics but not necessarily in bid distributions.

Each cartel chooses the cheapest available technology. In thick markets (large L), $c_w(L) > c_b$, and the cartel coordinates through bids: classical screens fire. In thin markets (small L), $c_w(L) < c_b$, and the cartel coordinates through personnel: worker-flow screens fire, but classical screens may not, because the bid distribution is not distorted by the coordination mechanism.

This framework generates three testable predictions:

1. **Direction**: In thin-labor markets, classical screen AUCs should be near chance; worker-flow AUCs should be above chance. The reverse holds in thick-labor markets.

2. **Placebo decomposition**: Because thin specialized-labor pools generate spontaneous inter-firm labor circulation even among non-colluding firms, a same-market placebo should exhibit elevated worker-flow overlap relative to random pairs (the specialization component). Cartel pairs should exhibit still greater overlap on top of that baseline (the coordination component). Both components should be present in thin-market data.
3. **Policy implication**: Enforcement strategies that rely exclusively on classical bid-pattern screens will systematically under-detect cartels in thin-labor environments, precisely where deterrence capacity may be weakest.

The model is deliberately stylized: it abstracts from dynamic considerations (repeated games, leniency incentives), from the possibility that firms use both technologies simultaneously, and from heterogeneity in c_b across auction formats. The framework is offered not as a structural model to be estimated but as an organizing device for interpreting our empirical findings and for specifying the conditions under which the classical-vs.-network screen comparison is most informative.

5. Data

Our analysis draws on three administrative data sources: the BEC procurement platform (bid-level records), the RAIS employer–employee matched panel (worker-flow construction), and CADE administrative proceedings (cartel ground truth). This section describes each source and the construction of the conviction-anchored benchmark samples.

5.1. BEC: bid-level procurement data

We use the universe of bid-level records from the *Bolsa Eletrônica de Compras* (BEC) platform for the period January 2009 to December 2019. The data comprise 3,866,407 individual bidding transactions submitted by 19,007 distinct firms across 169,607 unique procurement items. Each record identifies the purchase offer (PO) and item code, the participating firm (by CNPJ), the bid amount, the auction outcome (win or loss), the auction format (pregão or convite), and the identity of the purchasing agency. We restrict attention to competitive tenders with at least one winner and one loser, excluding canceled tenders and single-bid auctions. We define a unique auction as a (PO number $\times$ item code) pair, henceforth POI.

For each POI, we construct the auction-level variables described in Section 4.2: the coefficient of variation, bid spread, margin of victory, and number of active bidders, computed on firm-best bids (retaining each firm's lowest bid when multiple rounds exist). We define markets following Kawai et al. (2023): two firms belong to the same market if they participated at least once in the same auction for the same item code. This transitive market definition yields 7,581 distinct markets.

5.2. RAIS: employer–employee matched panel

The *Relação Anual de Informações Sociais* (RAIS) is the universe of formal-sector employment relationships in Brazil, collected annually by the Ministry of Labor. Every establishment with a CNPJ must file a RAIS declaration listing each employment bond active at any point during the calendar year, identified by the worker's PIS/PASEP number and the establishment's full 14-digit CNPJ. We use RAIS vinculos for the years 2009–2017, restricting to establishments whose 8-digit CNPJ root appears in the BEC firm universe (approximately 19,000 roots). The resulting panel contains all formal-sector employment bonds at BEC-active firms over the nine-year window.

From this panel, we construct the firm–firm worker-flow edge list described in Section 4.3. For each pair of BEC firms (i, j), we count the number of distinct PIS numbers that appear in the employment records of both firms at any point during the nine-year window (the shared-worker count W_{ij}) and compute the Jaccard similarity J_{ij}. We retain only edges with $W_{ij} \geq 1$; the remaining pairs are zero-imputed when joined to the auction-level benchmark.

5.3. CADE: cartel ground truth

The ground truth is constructed from CADE administrative proceedings with final Tribunal decisions convicting named firms of cartel conduct in São Paulo procurement markets (see Section 3 for institutional background). We identified seven relevant cartels spanning the sectors listed in Table 1, involving 35 firms with matchable CNPJ roots. For each cartel, the CADE case files specify a conduct period (start and end years) and the product market in which bid rigging or market division was found to have occurred.

We match each convicted firm's 8-digit CNPJ root to the BEC bidding records by firm identifier. We then label each BEC auction as *cartel-tainted* if at least one convicted firm participated during the CADE-specified conduct period for the relevant product market and year. This firm $\times$ period $\times$ product matching ensures that auctions are labeled as

cartel-tainted only when a convicted firm was bidding during the window in which its cartel was found to be active—not merely because the firm appears in a CADE database at some other time.

Table 1: Convicted Cartels in São Paulo Procurement Markets

Sector	CADE process	Period	Firms	In BEC	Conduct type
Trens e metrôs	08700.004617	1999–2013	13	10	Bid rigging
Merenda escolar	08012.010022	2006–2013	7	6	Market division
Medicamentos	08012.002222	2007–2011	8	8	Hub-and-spoke
Sacos de lixo	08700.005789	2008–2014	3	3	Bid rigging / price fixing
Aquecedores solares	08012.001273	2009–2010	5	4	Bid rigging
Cafeterias Infraero	08700.007278	2014	6	3	Bid rigging
Transporte escolar	08700.005876	2019	2	2	Bid rigging
Total (unique CNPJ roots)			35	36	

Notes: "Firms" counts unique 8-digit CNPJ roots per cartel; "In BEC" counts those matched to BEC bidding records. Total is after deduplication across cartels. Conduct periods are hand-curated from CADE Tribunal decisions and press releases (see Appendix for full audit). Source: script 12.

5.4. Sample construction

We construct three conviction-anchored detection samples, all restricted to the close-margin envelope $|MV| \leq 0.10$:

1. **Cartel-active benchmark** ($N \approx 2{,}300$ for pregão; $N = 263$ for convite): auctions in which at least one convicted firm participated during the active period of its cartel. This is the primary evaluation sample on which all screens are tested.
2. **Strict pair-matched sample** ($N = 10$ for pregão after close-margin filtering): auctions in which *both* the winner and the runner-up are firms convicted in the same cartel during the same conduct period. This sample is the gold standard for detecting coordination between known cartel partners but is severely power-limited.
3. **Anytime-listed sensitivity sample** ($N \approx 80$ for pregão): auctions in which both the winner and runner-up appear in a CADE conviction for the same cartel process, but the auction year may fall outside the prosecuted conduct window. Of the 107 pre-filtering auctions in this sample, 93 are out-of-window and

100 come from a single cartel (the medicamentos hub-and-spoke case, CADE process 08012.002222/2011-09). Because treating out-of-window bidding by former cartelists as collusive behavior requires assumptions we do not attempt to resolve, this sample is reported only as a sensitivity check.

The *clean control* consists of all close-margin auctions in which no cartel-listed firm participated ($N = 527{,}259$ for pregão; $N = 519{,}557$ for convite). All screens—classical and network-based—are evaluated against this common control on the unified close-margin benchmark.

5.5. Variable definitions

Table 2 summarizes the key variables used in the analysis.

Table 2: Variable Definitions

Variable	Definition
MV_a	Margin of victory: $(b_a^{\text{runner-up}} - b_a^{\text{win}})/b_a^{\text{win}}$
CV_a	Coefficient of variation of firm-best bids in auction a
S_a	Bid spread: $(\max - \min)/\text{mean}$ of firm-best bids
N_a	Number of distinct firms submitting bids in auction a
Incumbent_a	$= 1$ if the winning firm won the preceding auction in the same market
Last bidder_a	$= 1$ if the winning firm submitted the final bid
W_{ij}	Shared workers: count of distinct PIS appearing at both firms i and j
J_{ij}	Jaccard similarity: $W_{ij}/(N_i + N_j - W_{ij})$
C_{ij}	Co-bidding count: number of auctions where both i and j bid

6. Results

We present the empirical results in three parts. Section 6.1 reports the performance of classical bid-distribution screens and the close-margin RD on the unified benchmark. Section 6.2 reports the performance of worker-flow and co-bidding network screens. Section 6.3 reports the specialization-placebo decomposition.

6.1. Classical screens and the close-margin RD

Table 6 presents the unified benchmark comparison of all screens on the same close-margin pregão sample ($|MV| \leq 0.10$). Panel A reports the classical bid-distribution screens on the cartel-active benchmark (N = 2,319 cartel-tainted auctions vs. N = 527,259 clean controls).

All four classical screens fire in the direction predicted by the bid-rotation model: the coefficient of variation ($\Delta = -0.035$, $d = -0.07$), bid spread ($\Delta = -0.173$, $d = -0.10$), and number of active bidders ($\Delta = -0.87$, $d = -0.24$) are compressed at cartel-tainted auctions relative to clean controls. Welch t-statistics are large and statistically significant, owing to the massive control sample. Yet the effect sizes are small: Cohen's d ranges from -0.07 to -0.24, indicating near-total distributional overlap between cartel and clean auctions. The AUCs reflect this overlap, ranging from 0.52 (coefficient of variation) to 0.56 (number of bidders)—above chance but far too low for investigative triage. A Wilcoxon rank-sum test confirms the weak rank-order separation ($p < 10^{-4}$ for each screen, consistent with the AUC direction).

The margin of victory exhibits an anomalous pattern: $\Delta MV = +0.004$ ($d = +0.17$), with an AUC of 0.46—slightly *below* chance and in the *opposite* direction to the bid-rotation prediction. Cartel-active auctions have marginally *larger* winning margins, not smaller. We interpret this as consistent with a partial phantom-bidding mechanism in which cover bids are inflated relative to the designated winner's price, widening rather than compressing the margin. This finding warrants further investigation but does not affect the headline conclusion: classical screens as a portfolio do not achieve operational discriminative power on this benchmark.

A direct replication of the Kawai et al. (2023) close-margin RD on the cartel-active subsample yields no statistically detectable discontinuity in incumbency ($\hat{\beta} = 0.018$, SE = 0.005 on the full sample; $p > 0.15$ on the cartel-active subsample) or last-bid behavior at the cutoff. A density test in the spirit of Imhof et al. (2018) detects no manipulation of the running variable ($T_{\text{CJM}} \approx 0$, $p = 1.0$ on the cartel-active benchmark). Neither the RD design nor the density test flags the cartel-active auctions as anomalous relative to competitive bidding.

6.2. Network-based screens

Panel B of Table 6 reports the network screens on the same unified benchmark (N = 2,319 cartel-active pregão auctions, $|MV| \leq 0.10$).

The worker-flow screens achieve markedly higher discriminative power. The shared-worker count yields an AUC of 0.750 ($\Delta = +3.70$, $t = +14.1$), and the Jaccard similarity yields an AUC of 0.739 ($\Delta = +0.0014$, $d = +0.19$, $t = +20.7$). Both outperform the best classical screen (number of bidders, AUC = 0.56) by roughly 20 percentage points. A third screen constructed from co-bidding density yields an AUC of 0.695, providing independent corroboration through a distinct information channel.

Critically, the worker-flow signal is not an artifact of sectoral composition. Restricting attention to firm pairs that share the same 2-digit CNAE sectoral classification reduces the Jaccard AUC only marginally, from 0.739 to approximately 0.730. The residual of worker-flow after partialling out the log co-bidding density of each pair achieves an AUC of 0.749—essentially unchanged from the raw baseline, indicating that worker mobility carries detection information above and beyond what co-bidding patterns alone can reveal.

On the convite subsample ($N = 263$ cartel-active, $|MV| \leq 0.10$), the worker-flow screens maintain strong performance: shared-worker AUC = 0.704, Jaccard AUC = 0.701. The consistency of network-screen performance across auction formats—in contrast to the format-sensitive behavior of classical screens—supports the view that worker-flow connectivity captures a channel of coordination that is invariant to the procedural details of the bidding mechanism.

A natural question is what the cartel-active AUC actually measures, given that in most auctions only *one* of the two firms in the (winner, runner-up) pair is a convicted cartelist. Of the 2,319 cartel-active auctions, 996 have the *winner* as the cartel-active firm and the runner-up as a non-cartel firm; 1,313 have the *runner-up* as the cartel-active firm; and only 10 have both firms cartel-active. Decomposing the AUC by cartel role clarifies the interpretation:

Subgroup	N	AUC (shared workers)	AUC (Jaccard)
Winner cartel, runner-up clean	996	0.741	0.730
Runner-up cartel, winner clean	1,313	0.759	0.747
Both cartel-active	10	0.558	0.550

Three observations emerge. *First*, the AUC is remarkably stable across the two one-sided subgroups (0.74–0.76), indicating that the detection signal is symmetric: it does not matter whether the cartel firm wins or loses the auction. This rules out the concern

that the screen merely captures a "winner effect" driven by a cartel firm's incumbency or market dominance. *Second*, the signal is present even when the *runner-up* is the cartel firm (AUC = 0.759), confirming that elevated worker-flow connectivity is a property of cartel-firm pairs in thin labor markets, not an artifact of winning per se. *Third*, intra-cartel pairs (both firms cartel-active, $N = 10$) exhibit the highest mean shared-worker count ($\bar{W} = 28.4$, compared with 6.5 for winner-only and 2.1 for clean controls) but an AUC near chance (0.56), reflecting severe power constraints at small N rather than an absence of signal.

The decomposition clarifies the mechanism: the worker-flow screen detects that cartel firms are embedded in thin specialized-labor markets where personnel flows with *all* rivals—not just co-conspirators—are dense. The specialization placebo (Section 6.3) further separates this market-embeddedness component from the cartel-specific coordination residual. The headline AUC of 0.75 thus captures both channels; the cartel-specific residual, net of market specialization, is approximately 0.64 (Section 6.3).

6.3. Specialization-placebo decomposition

Table 7 reports the three-way comparison of worker-flow distributions across the cartel, same-market placebo, and clean-global samples (pregão, $|MV| \leq 0.10$).

Cartel pairs share at least one worker in 57.9% of cases, compared with 31.1% for same-market placebo pairs and 9.9% for pairs in other markets. The Jaccard AUC against the global clean baseline is 0.740 for cartel pairs and 0.605 for same-market placebo pairs. In a linear decomposition of the total AUC gain (cartel vs. clean minus 0.5):

- The **specialization component** (placebo vs. clean) absorbs approximately 0.105 AUC points, or roughly 44% of the total signal.
- The **cartel-specific residual** (cartel vs. placebo) accounts for approximately 0.138 AUC points, or roughly 56% of the total signal.

Both components are consistent with the institutional-thinness mechanism of Section 4.6: thin specialized-labor markets generate spontaneous labor circulation among same-market firms (the specialization component), and cartels additionally reinforce those flows to sustain coordination (the cartel-specific residual). The two-component

structure also has a practical implication: a worker-flow screen deployed without conditioning on market identity would capture both components, potentially flagging non-colluding same-market firm pairs as suspicious. Conditioning on market or product code absorbs the specialization component, yielding a purer cartel signal at the cost of a modest reduction in overall AUC (from 0.740 to 0.638).

6.4. Heterogeneity by market thickness

The institutional-thinness model of Section 4.6 predicts that worker-flow screens should outperform classical screens most sharply in thin markets, where the cheapest coordination technology operates through personnel flows rather than bid-pattern manipulation. We test this prediction by splitting the cartel-active benchmark according to market thickness, measured as the median number of bidders per auction for each item code in the clean control sample. Markets with a control-sample median at or below the overall median of 5 firms are classified as "thin"; those above are classified as "thick." This proxy is constructed entirely from the non-cartel sample to avoid contamination by cartel-induced changes in bidder counts.

Table 3 reports the results. The pattern is striking. In thin markets ($N_{\text{cartel}} = 1{,}618$), classical screens fire in the *wrong* direction: CV AUC = 0.45, spread AUC = 0.44, and bidder-count AUC = 0.40—all below chance. Cartel-tainted auctions in thin markets exhibit *more* bid dispersion and *more* active bidders than clean controls, consistent with a phantom-bidding or cover-bid mechanism in which all cartel members participate and submit inflated bids. By contrast, in thick markets ($N_{\text{cartel}} = 416$), classical screens fire in the predicted direction: CV AUC = 0.59, spread = 0.60, bidder count = 0.60—substantively above chance and consistent with the standard bid-rotation model.

Worker-flow screens show the opposite gradient. In thin markets, shared-worker AUC reaches 0.774 and Jaccard AUC reaches 0.758; in thick markets, both decline to approximately 0.70. The gap between the best classical screen and the worker-flow screen is 33 percentage points in thin markets but only 10 percentage points in thick markets—a three-fold difference that confirms the model's core prediction.

The tercile decomposition (reported in the notes to Table 3) reveals additional structure: the worker-flow AUC peaks in mid-thickness markets (tercile 2: AUC = 0.80) rather than in the thinnest markets (tercile 1: AUC = 0.75). This non-monotonicity is consistent with the specialization-placebo finding of Section 6.3: in the very thinnest markets, spontaneous same-market labor circulation is so dense that it partially com-

presses the cartel-vs-clean AUC from above, even as the cartel-specific signal remains strong.

Table 3: Screen Performance by Market Thickness

	N_{cartel}	N_{clean}	CV	Spread	N_{firms}	Shared workers	Jaccard
Thin (≤ 5 firms)	1,618	217,662	0.447	0.436	0.400	**0.774**	**0.758**
Thick (> 5 firms)	416	224,074	0.594	0.604	0.604	0.699	0.694
All (benchmark)	2,319	527,259	0.523	0.531	0.555	**0.750**	**0.739**

Notes: Market thickness is the median number of bidders per auction for each item code, computed on clean-control auctions with ≥ 5 observations. "Thin" = median ≤ 5; "Thick" = median > 5. Classical screen AUCs use "lower is cartel" convention. Tercile decomposition: T1 (median = 4): $N_t = 1{,}191$, shared-workers AUC = 0.751, CV AUC = 0.382; T2 (median = 5): $N_t = 602$, shared-workers AUC = 0.797, CV AUC = 0.533; T3 (median = 8): $N_t = 241$, shared-workers AUC = 0.644, CV AUC = 0.612.

6.5. Model predictions vs. empirical findings

Table 4 summarizes the correspondence between the three testable predictions of the institutional-thinness model (Section 4.6) and the empirical results documented in this section.

Table 4: Model Predictions and Empirical Results

Prediction	Empirical test	Result
P1. In thin markets, classical AUCs ≈ 0.5; worker-flow AUCs > 0.5.	Table 3: median split by market thickness	Classical AUC = 0.40–0.45 (inverted); worker-flow AUC = 0.77. **Confirmed.**
P2. Same-market placebo exhibits elevated worker-flow relative to random pairs; cartel pairs exhibit still more.	Table 7: three-way decomposition	Specialization component = 44%; cartel residual = 56%. Both present. **Confirmed.**
P3. Classical enforcement under-detects cartels where deterrence value is highest.	Table 3: gap thin vs. thick	Gap = 33 pp in thin markets; 10 pp in thick. **Confirmed.**

All three predictions are borne out in the data. The model's qualitative framework—that the cheapest coordination technology determines which screen family carries the detection signal—is thus not merely an organizing device but a tested empirical proposition, at least within the institutional setting of São Paulo procurement.

6.6. Selection and the interpretation of the benchmark

A natural concern with any conviction-anchored benchmark is that the cartels CADE detected and convicted are not a random sample of all cartels that existed during the study period. If the detection channel itself correlates with worker-flow intensity—for instance, if CADE is more likely to investigate firms when their personnel connections are visible to investigators—then the worker-flow AUC may overstate the screen's performance on the universe of undetected cartels. We address this concern in three ways.

First, we document the evidentiary basis of each of the seven convicted cartels from the public record of CADE's administrative proceedings. Table 5 summarizes the primary detection channel and the nature of the evidence that sustained each conviction. The evidence channels are heterogeneous: one case (aquecedores solares) was convicted "based exclusively on indirect evidence"—that is, bid-pattern analysis alone; one case (transporte escolar) relied on IP address coincidence during electronic bidding; one case (sacos de lixo) originated from a state-level police operation (*Operação Colludium*); and the remaining four were built from leniency confessions, procurement-document analysis, or investigation-triggered dawn raids. Crucially, *none* of the seven cases mentions matched employer–employee data, RAIS records, or inter-firm personnel connections as part of the evidentiary basis for the investigation or conviction. CADE does not, as a matter of standard practice, use worker-flow analysis as an investigative tool—the data linkage we construct in this paper is, to our knowledge, the first of its kind. The detection channel (CADE investigations driven by complaints, leniency, police operations, and bid-pattern screens) is therefore independent of the screening channel (worker-flow connectivity measured from RAIS).

Second, we note that the worker-flow screen is constructed from an entirely external administrative data source (RAIS, held by the Ministry of Labor) that is not part of CADE's standard enforcement toolkit. Unlike co-bidding patterns—which are constructed from the same procurement data that CADE analysts routinely examine—worker-flow connectivity requires a separate data-access agreement and a non-trivial linkage procedure. The informational independence of the two channels reduces the scope for mechanical circularity.

Third, we observe that the worker-flow AUC is relatively stable across the seven cartels despite the heterogeneity in detection channels. If selection-on-detection were driving the result, we would expect the worker-flow signal to be concentrated in cases

where personnel connections played a role in the investigation. Instead, the signal is present even in the aquecedores solares case—convicted on bid-pattern evidence alone—and in the transporte escolar case—detected through IP forensics. While this cross-cartel stability is suggestive rather than definitive (small N of cartels limits formal testing), it is inconsistent with a pure selection story.

These observations do not fully eliminate the selection concern. It remains possible that firms with dense worker-flow networks are easier to detect through *any* channel—for instance, because the same organizational proximity that generates personnel flows also generates careless document trails or complaints from disgruntled employees. A definitive test would require evaluating the worker-flow screen against cartels detected through channels that are *provably* orthogonal to personnel connectivity, such as pure leniency applications from firms with no prior labor-market overlap. We leave this test to future work and flag the selection caveat as the principal interpretive limitation of our benchmark design.

7. Discussion and Conclusion

This paper assembles a conviction-anchored benchmark—seven CADE-convicted procurement cartels linked to 3.8 million bids and the universe of formal employment in São Paulo—and uses it to ask which detection tools actually work. Three findings stand out.

The first is a warning. Canonical variance screens behave as bid-rotation theory predicts—dispersion and bidder counts are compressed at cartel-tainted auctions—but the effect sizes are small and the distributional overlap with clean auctions is near-total. AUCs top out at 0.56. The close-margin RD of Kawai et al. (2023) produces a null. Neither design is internally flawed; both simply lack the statistical punch to separate cartels from noise in this institutional setting.

The second finding is constructive. Worker-flow screens built from matched employer–employee records achieve AUCs of 0.75, outperforming the best classical screen by roughly 20 percentage points. A co-bidding density measure reaches 0.70. A specialization placebo attributes about half the worker-flow signal to the natural personnel circulation of thin product markets and the other half to cartel-specific coordination. The signal is symmetric across winner and runner-up roles, survives conditioning on sector and co-bidding intensity, and is stable across bandwidths.

The third finding is the one that ties the story together. The gap between classical

Table 5: Evidence Channels in the Seven Convicted Cartels

Cartel (sector)	CADE process	Period	Primary evidence channel
Merenda escolar	08012.010022/ 2008-16	2006–2013	Procurement documents (40,000+ docs analyzed); geographic market division
Trens e metrôs	08700.004617/ 2013-41	1999–2013	Leniency and investigation; subcontracting arrangements between winners and losers
Aquecedores solares	08012.001273/ 2010-24	2009–2010	Indirect evidence only (bid-pattern analysis); no direct evidence of agreement
Transporte escolar	08700.005876/ 2019-85	2019	IP address coincidence during electronic bidding
Cafeterias Infraero	08700.007278/ 2015-17	2014	Investigation of in-person tender procedures
Sacos de lixo	08700.005789/ 2015-02	2008–2014	Police operation (*Op. Colludium*, MP Bauru); dawn raids and document seizures
Medicamentos	08012.002222/ 2011-09	2007–2011	Hub-and-spoke investigation; leniency and administrative proceedings

Notes: Evidence channels documented from CADE public press releases and Tribunal decisions (see Appendix for full citations). None of the seven cases lists matched employer–employee data, RAIS records, or inter-firm worker flows as part of the evidentiary basis.

Table 6: Unified Benchmark: All Screens on the Same Close-Margin Sample

	Pregão ($N_t = 2{,}319$)		Convite ($N_t = 263$)	
Screen	Cohen's d	AUC	Cohen's d	AUC
Panel A: Classical bid-distribution screens				
Coefficient of variation (CV)	−0.07	0.523	−0.01	0.515
Bid spread (S)	−0.10	0.531	+0.04	0.502
Margin of victory (MV)	+0.17	0.457	+0.14	0.456
Number of bidders (N)	−0.24	0.555	+0.06	0.467
Panel B: Network-based screens				
Shared workers (W_{ij})	+0.06	**0.750**	+0.47	**0.704**
Jaccard similarity (J_{ij})	+0.19	**0.739**	+0.09	**0.701**
Co-bidding density (C_{ij})	+0.14	0.695	−0.04	0.525

Notes: All screens evaluated on the unified close-margin benchmark ($|MV| \leq 0.10$) against a common clean control ($N_0 = 527{,}259$ pregão; $N_0 = 519{,}557$ convite). Classical screen AUCs use the "lower is cartel" convention; values above 0.5 indicate detection in the predicted direction. Network screen AUCs use "higher is cartel." Bold indicates AUC ≥ 0.70. Cohen's d is the standardized mean difference (pooled SD). Source: script 23.

Table 7: Specialization-Placebo Decomposition of Worker-Flow Signal

Sample	N	% with ≥ 1 shared worker	Mean Jaccard	AUC vs. clean
Cartel-active	2,321	57.9%	0.0022	0.740
Placebo (same market)	5,313	31.1%	0.0012	0.605
Clean (global)	523,414	9.9%	0.0008	—
Signal decomposition (Jaccard AUC relative to clean baseline)				
Total signal (cartel vs. clean − 0.5)				+0.240
Specialization component (placebo vs. clean)				+0.105 (44%)
Cartel-specific residual (cartel vs. placebo)				+0.138 (56%)

Notes: Pregão, $|MV| \leq 0.10$. Placebo consists of non-cartel firm pairs bidding on the same BEC item code in the same year as each cartel-active auction. AUC computed via Mann–Whitney U (Jaccard similarity, higher is cartel). Percentages in decomposition are shares of total signal. Source: v3 specialization placebo.

and worker-flow screens is not constant: it widens from 10 percentage points in thick labor markets to 33 in thin ones. In thin markets, classical screens actually fire in the wrong direction—cartel auctions show *more* dispersion, not less. This gradient confirms the institutional-thinness mechanism we develop in Section 4.6: where specialized-labor pools are shallow, the cheapest coordination technology runs through personnel rather than prices, and the detection signal migrates from bid patterns to employment records.

Policy implications. For competition authorities in middle-income economies, the message is direct. Matched employer–employee registries—RAIS in Brazil, DADS in France, IAB in Germany—are increasingly available and digitized. Building a worker-flow screen on top of existing procurement data requires a one-time linkage, not a new data-collection program. The marginal cost is negligible; the detection gain, in thin markets, is large. Authorities that rely exclusively on imported variance screens are not just under-detecting cartels—they are under-detecting them precisely in the thinnest, most vulnerable markets, where deterrence would matter most.

Limitations. The principal caveat is selection. CADE convictions are not a random sample of all cartels, and although none of the seven cases in our benchmark was built on worker-flow evidence (Section 6.6), we cannot fully rule out that firms with dense personnel networks are easier to catch through any channel. A definitive test would require ground truth from a detection process provably orthogonal to labor-market connectivity—pure leniency applications from firms with no prior worker overlap, for instance. The strict pair-matched sample ($N = 10$) is too small to draw conclusions about head-to-head coordination; larger ground-truth sets, assembled from cross-state CADE linkages, would sharpen the test. And the simple model of Section 4.6 treats bid-pattern and worker-flow coordination as substitutes; a richer framework allowing complementarity—where cartels use both channels at once—could generate additional predictions and refine the screen-design implications.

Looking ahead. The broader lesson is that collusion is not one thing. It adapts to the institutional soil it grows in, and the tools built to detect it must adapt too. A screen calibrated for thick, well-policed markets may be exactly the wrong instrument for thin ones—not because it is poorly designed, but because it is listening for the wrong signal. Following the workers, rather than the bids, is one way to recalibrate. There are likely others.

References

Abrantes-Metz, R. M., L. M. Froeb, J. F. Geweke, and C. T. Taylor (2006). A variance screen for collusion. *International Journal of Industrial Organization 24*(3), 467–486.

Asai, K. and B. Charoenwong (2026). Ownership networks and bid rigging. *The Review of Corporate Finance Studies*. Advance article, published online February 2026.

Bajari, P. and I.-K. Ye (2003). Detecting collusion in procurement auctions. *Journal of Law, Economics, & Organization 19*(2), 453–492.

Barbosa, K. and S. Straub (2020). The value of revolving doors in public procurement. Working Paper 17-873, Toulouse School of Economics. R&R at Journal of Law, Economics, and Organization.

Bosio, E., S. Djankov, E. Glaeser, and A. Shleifer (2022). Public procurement in law and practice. *American Economic Review 112*(3), 1091–1117.

Chassang, S. and J. Ortner (2019). Collusion in auctions with constrained bids: Theory and evidence from public procurement. *Journal of Political Economy 127*(5), 2269–2300.

Clark, R., D. Coviello, J.-F. Gauthier, and A. Shneyerov (2018). Bid rigging and entry deterrence in public procurement: Evidence from an investigation into collusion and corruption in Québec. *Journal of Law, Economics, and Organization 34*(3), 301–363.

Conley, T. G. and F. Decarolis (2016). Detecting collusive behavior in public procurement auctions: Evidence from competitive bidding. *Journal of Public Economics 143*, 1–13.

Harrington, Joseph E., J. (2008). Detecting cartels. In P. Buccirossi (Ed.), *Handbook of Antitrust Economics*, pp. 213–258. Cambridge, MA: MIT Press.

Herrera-Caicedo, A., J. Jeffers, and E. Prager (2025). Collusion through common leadership. Working Paper 33866, National Bureau of Economic Research. Accepted, Journal of Political Economy.

Huber, M. and D. Imhof (2019). Machine learning with screens for detecting bid-rigging cartels. *International Journal of Industrial Organization 65*, 277–301.

Imhof, E., L. Musolff, and V. Titl (2018). Screening for collusion in public procurement auctions. *European Journal of Political Economy 55*, 47–62.

Jarosch, G., E. Oberfield, and E. Rossi-Hansberg (2021). Learning from coworkers. *Econometrica 89*(2), 647–676.

Kawai, K., J. Nakabayashi, J. Ortner, and S. Chassang (2023). Using bid rotation and

incumbency to detect collusion: A regression discontinuity approach. *The Review of Economic Studies 90*(1), 376–403.

Lamichhane, K. and P. McGee (2024). Collusion through worker movement: An experiment. Job Market Paper, University of Arkansas.

Levenstein, M. C. and V. Y. Suslow (2006). What determines cartel success? *Journal of Economic Literature 44*(1), 43–95.

Marshall, R. C. and L. M. Marx (2012). *The Economics of Collusion: Cartels and Bidding Rings*. Cambridge, MA: MIT Press.

Organisation for Economic Co-operation and Development (2015). *Government at a Glance 2015*. Paris: OECD Publishing.

Pesendorfer, M. (2000). Study of collusion in school milk auctions: Evidence from florida and texas. *Journal of Law, Economics, & Organization 16*(1), 31–50.

Porter, R. H. and D. A. Zona (1993). Detecting collusive behavior in procurement auctions. *Journal of Law, Economics, & Organization 9*(1), 157–194.

Serafinelli, M. (2019). "good" firms, worker flows, and local productivity. *Journal of Labor Economics 37*(3), 747–792.

Signor, R., P. E. D. Love, A. T. N. Belarmino, and O. A. Olatunji (2020). Detection of collusive tenders in infrastructure projects: Learning from Operation Car Wash. *Journal of Construction Engineering and Management 146*(1), 05019015.

Stoyanov, A. and N. Zubanov (2012). Productivity spillovers across firms through worker mobility. *American Economic Journal: Applied Economics 4*(2), 168–198.

Szerman, C. (2023). The employee costs of corporate debarment in public procurement. *American Economic Journal: Applied Economics 15*(1), 411–441.

Villamil, I., J. Kertész, and M. Fazekas (2024). Collusion risk in corporate networks. *Scientific Reports 14*, 4006.

Wachs, J. and J. Kertész (2019). A network approach to cartel detection in public auction markets. *Scientific Reports 9*(1), 10818.

A. Appendix

A.1. Bandwidth sensitivity

Table A.1 reports AUCs for all screens across six bandwidth choices for the close-margin envelope. The benchmark bandwidth of $|MV| \leq 0.10$ used throughout the paper is neither the most nor the least favorable for the worker-flow screens. At narrower bandwidths (0.02, 0.05), worker-flow AUCs are marginally higher (0.76, 0.75) but sample sizes are smaller; at wider bandwidths (0.20, 0.50), AUCs decline modestly (0.74, 0.72) as the sample dilutes toward the full population. Classical screen AUCs follow the same monotonic pattern. The gap between the best classical screen (number of bidders) and the worker-flow screens remains stable at approximately 18–20 percentage points across all bandwidths.

Table A.1: AUC Sensitivity to Close-Margin Bandwidth (Pregão, Cartel-Active)

$\|MV\| <$	N_{cartel}	N_{clean}	CV	Spread	N_{firms}	Shared workers	Jaccard
0.02	989	277,038	0.546	0.558	0.580	0.758	0.747
0.05	1,700	416,582	0.539	0.548	0.565	0.760	0.749
0.10	**2,319**	**527,259**	**0.523**	**0.531**	**0.555**	**0.750**	**0.739**
0.15	2,607	573,196	0.518	0.526	0.554	0.745	0.734
0.20	2,784	602,883	0.517	0.525	0.555	0.743	0.732
0.50	3,277	693,331	0.518	0.526	0.552	0.735	0.723

Notes: Each row reports AUCs from the unified benchmark (script 23) at the indicated bandwidth. Classical screen AUCs use the "lower is cartel" convention (negated scores). Bold row indicates the benchmark bandwidth used in the main analysis. All AUCs computed via Mann–Whitney U with average-rank tie-handling.